CareFlow – AI Care Coordination Assistant

Project ID: 11

1-Month Project Final Progress

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GitHub Repository: CareFlow on GitHub

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1. Project Overview & 1-Month Summary

CareFlow is an intelligent, multi-agent AI clinical triage and care coordination platform designed to automate patient intake, perform structured clinical disease assessment, enforce medical safety protocols, retrieve verified clinical care guidelines (RAG), and book appointments with specialized physicians while keeping a human clinician in the loop (HITL).

Over the four-week project lifecycle, CareFlow evolved through:

1. **Week 1 (Understanding & Requirements):** Formulated the clinical problem statement, care coordination bottlenecks, and safety scope.
2. **Week 2 (System Design & Architecture):** Architected the modular LangGraph state machine, typed schemas (`CareFlowState`), RAG pipeline, and MCP tools.
3. **Week 3 (Implementation & Prototype Demonstration):** Developed the full-stack FastAPI application, LangGraph state machine, 30+ condition database, and Clinician HITL Desk.
4. **Week 4 (Evaluation, Slips & Final Presentation):** Conducted a 25-case empirical evaluation against a monolithic LLM baseline, built automated ReportLab PDF Clinical Referral Slip generation, added live evaluation telemetry, and developed the presentation package for October 16th.

2. Key Capabilities & Empirical Benchmarks

In empirical evaluations across 25 representative clinical test cases (`benchmark_evaluation.py`), CareFlow achieved:

- **100.0% Emergency Red-Flag Recall** (compared to 72.0% in monolithic LLMs).
- **100.0% Medication Safety & Contraindication Adherence** (preventing dangerous NSAID/Aspirin recommendations in dengue and pediatric presentations).
- **6.24 ms Average Triage Latency** using an asynchronous non-blocking offline heuristic engine.
- **100.0% Pydantic Schema Determinism** ensuring end-to-end auditability and seamless EHR interoperability.
- **Automated One-Click Clinical Referral Slip Generation** exporting verified PDF consultation tokens (`BK-1001`) with clinician sign-off.

3. How to Run the Application & Prototype

3.1 Prerequisites

- Python 3.10+ (the supplied text says tested up to Python 3.14).
- Create and activate a virtual environment, then install dependencies:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

3.2 Step 1: Start the CareFlow Server

```
source .venv/bin/activate
python server.py
```

The server runs at http://localhost:8000.

3.3 Step 2: Open the Web Portal

Open a browser and navigate to http://localhost:8000.

3.4 Step 3: Run the Empirical Benchmark Suite

To execute the automated 25-case clinical evaluation comparing CareFlow against a monolithic LLM:

```
source .venv/bin/activate
python benchmark_evaluation.py
```

This prints the formatted evaluation table and updates `benchmark_results.json`.

3.5 Step 4: View the Final Presentation Deck

Open `CareFlow_Presentation.html` in a browser:

```
xdg-open CareFlow_Presentation.html
```

Alternatively, open it through the local server or double-click the HTML file.

- Navigate slides using the Left (←) and Right (→) arrow keys or the on-screen buttons.
- Review speaker notes and anticipated mentor Q&A in `CareFlow_Final_Presentation.md`.

4. Verification & Demonstration Flow for Mentors

4.1 Patient Registration Modal

On initial launch, enter patient details, for example: **Rohan Sharma**, Age: **28**, City: **Hyderabad**, Condition: **Fever**.

4.2 Clinical Disease Assessment

In the chat, type queries such as:

- "I have a fever, what precautions and medicine should I take?"
- "What are the symptoms and precautions for Dengue?"
- "I have severe migraine headaches."

Observe the structured six-part clinical response:

1. Clinical Symptoms
2. Home Care & Precautions
3. Safe OTC Medications with Limits
4. What to Avoid (Contraindications)
5. Red-Flag Emergency Alerts
6. Confirmed Doctor Appointment Slot

4.3 Emergency Red-Flag Screening

Type "Sudden numbness in right arm and slurred speech" or "Crushing chest pain and breathlessness." The intended demonstration is for the Safety Agent to trigger an emergency 911/112 protocol, halt routine scheduling, and prioritize emergency medical transfer.

4.4 Clinician Oversight (HITL Desk)

Click the **Clinician HITL Desk** tab at the top. Review pending cases, click **Approve & Sign-off**, and view clinical history.